MS-13 · KPM DROPPER Ph-2

Code 910 | Answer Key & Detailed Solutions

PART 1: ANSWER KEY – PHYSICS (1–45)

Q1.4	Q2.4	Q3.4	Q4.4	Q5.2	Q6.3	Q7.3	Q8.3	Q9.1	Q10.4
Q11.3	Q12.4	Q13.4	Q14.4	Q15.2	Q16.2	Q17.4	Q18.3	Q19.2	Q20.4
Q21.4	Q22.2	Q23.4	Q24.3	Q25.3	Q26.3	Q27.4	Q28.1	Q29.1	Q30.4
Q31.3	Q32.1	Q33.2	Q34.2	Q35.2	Q36.4	Q37.2	Q38.3	Q39.1	Q40.2
Q41.3	Q42.3	Q43.2	Q44.3	Q45.3					

CHEMISTRY (46–90)

Q46.2	Q47.1	Q48.4	Q49.1	Q50.2	Q51.2	Q52.1	Q53.4	Q54.4	Q55.3
Q56.2	Q57.2	Q58.1	Q59.2	Q60.3	Q61.3	Q62.4	Q63.4	Q64.3	Q65.2
Q66.2	Q67.3	Q68.4	Q69.4	Q70.3	Q71.2	Q72.2	Q73.4	Q74.1	Q75.4
Q76.3	Q77.2	Q78.4	Q79.2	Q80.2	Q81.3	Q82.3	Q83.4	Q84.3	Q85.3
Q86.1	Q87.4	Q88.4	Q89.2	Q90.4					

BIOLOGY (91–180)

Q91.3	Q92.3	Q93.2	Q94.3	Q95.4	Q96.4	Q97.1	Q98.4	Q99.4	Q100.4
Q101.4	Q102.1	Q103.2	Q104.3	Q105.3	Q106.1	Q107.2	Q108.3	Q109.3	Q110.1
Q111.3	Q112.4	Q113.3	Q114.4	Q115.4	Q116.4	Q117.2	Q118.3	Q119.2	Q120.1
Q121.3	Q122.3	Q123.1	Q124.3	Q125.3	Q126.3	Q127.2	Q128.3	Q129.2	Q130.2
Q131.3	Q132.3	Q133.2	Q134.1	Q135.3	Q136.2	Q137.3	Q138.3	Q139.4	Q140.2
Q141.4	Q142.4	Q143.3	Q144.3	Q145.2	Q146.4	Q147.3	Q148.2	Q149.4	Q150.4
Q151.2	Q152.2	Q153.2	Q154.1	Q155.3	Q156.4	Q157.4	Q158.2	Q159.2	Q160.2
Q161.3	Q162.4	Q163.3	Q164.4	Q165.4	Q166.2	Q167.2	Q168.2	Q169.4	Q170.3
Q171.4	Q172.1	Q173.1	Q174.3	Q175.2	Q176.1	Q177.3	Q178.1	Q179.2	Q180.3

PART 2: SOLUTIONS

PHYSICS

1. $v = 0$ ho to $F = qvB = 0$, kan rest mein rahega. **Ans. 4**

2. Loop ke andar $i + 3i + i = 5i$ ($2i$ bahar hai). $\oint \vec{B} \cdot d\vec{l} = 5\mu_0 i$. **Ans. 4**

3. $M = nI\pi r^2 \propto r^2$. **Ans. 4**

4. B, current ke parallel hai, isliye $F = BIl\sin 0^\circ = 0$. **Ans. 4**

5. $\chi < 0$ sirf diamagnetic ke liye. **Ans. 2**

6. Domain sirf ferromagnetism mein bante hain. **Ans. 3**

7. Diamagnetic: $\chi < 0$, $\mu_r < 1$. **Ans. 3**

8. $B_{end} = \frac{1}{2}B_{centre}$, isliye ratio 2 : 1. **Ans. 3**

9. $\chi = \mu_r - 1 = 22$. **Ans. 1**

10. Gauss law: monopole nahi hota aur field lines closed loop hoti hain. **Ans. 4**

11. Ferromagnetic ki permeability sabse zyada. **Ans. 3**

12. A, B, C teeno sahi. **Ans. 4**

13. Small magnet ka field $B \propto \frac{1}{R^3}$. **Ans. 4**

14. Copper diamagnetic hai, isliye repel hota hai. **Ans. 4**

15. Effective length $= 2R$, $F = 2BIR$. **Ans. 2**

16. $T = \frac{2\pi m}{qB}$, to half revolution $= \frac{\pi m}{qB}$. **Ans. 2**

17. $B_0 = \frac{\mu_0 i}{2R}$. Axial point ($x = \sqrt{3}R$) par $B = \frac{\mu_0 i R^2}{2(4R^2)^{3/2}} = \frac{\mu_0 i}{16R}$. Ratio 8 : 1. **Ans. 4**

18. Dono statement sahi ($v \parallel B$ to $F = 0$). **Ans. 3**

19. Diamagnetic χ, T par depend nahi karta, isliye horizontal line. **Ans. 2**

20. $\frac{F}{L} = Bi\sin\theta = 2 \times 10^{-3} \times 4 \times \frac{1}{2} = 4 \times 10^{-3}$ N/m. **Ans. 4**

21. $W = U_f - U_i = MB - (-MB) = 2MB$. **Ans. 4**

22. $d = R\sin\theta$ aur $d = \frac{\sqrt{3}}{2}R$, to $\sin\theta = \frac{\sqrt{3}}{2}$. **Ans. 2**

23. Bahar field N→S, andar S→N (upar ki taraf). **Ans. 4**

24. Beech se todne par pole strength m wahi rehti hai aur length aadhi, isliye $\frac{M}{2}$, m. **Ans. 3**

25. i_1 aur i_2 ke fields equal aur opposite hain, net zero. **Ans. 3**

26. Dia: $0 > \chi \geq -1$, Ferro: $\chi \gg 1$, Para: chhota positive, Non-magnetic: $\chi = 0$. A-II, B-III, C-IV, D-I. **Ans. 3**

27. $B = \mu_0 nI$, I doguna aur n aadha, to B same. **Ans. 4**

28. Curie temperature ke upar ferro, para ban jata hai. **Ans. 1**

29. A aur R dono sahi, R sahi explanation. **Ans. 1**

30. $d\vec{B} = \frac{\mu_0}{4\pi}\frac{i(d\vec{l} \times \vec{r})}{r^3}$. **Ans. 4**

31. Andar $B \propto d$, bahar $B \propto \frac{1}{d}$. **Ans. 3**

32. $M = IA = \frac{q\omega}{2\pi}(\pi R^2) = \frac{q\omega R^2}{2}$. **Ans. 1**

33. $\vec{M}_1$ $-x$ direction mein (180°), $\vec{M}_2$ $+x$ se 60° par. Beech ka angle 120°. **Ans. 2**

34. $r = \frac{p}{qB} \propto \frac{1}{q}$, $q_\alpha : q_d = 2 : 1$, to $r_d : r_\alpha = 2 : 1$. **Ans. 2**

35. $T = \frac{2\pi m}{qB}$, angle par depend nahi karta. **Ans. 2**

36. $B = \mu_r B_0$, to $\mu_r = 4$. **Ans. 4**

37. $B = \frac{\mu_0 i}{4\pi r}(\sin 53^\circ - \sin 37^\circ) = \frac{\mu_0 i}{20\pi r}$. Direction RHR se andar ki taraf ($\otimes$). **Ans. 2**

38. $r^2 = \frac{2mV}{qB^2} \propto m$, isliye $\frac{m_X}{m_Y} = \left(\frac{R_1}{R_2}\right)^2$. **Ans. 3**

39. $M = ml$. Arc length $l = \frac{\pi R}{3}$, chord $= R$. $M' = mR = \frac{3M}{\pi}$. **Ans. 1**

40. Magnetic force kaam nahi karta, $KE = K$. **Ans. 2**

41. Andar $B = \frac{\mu_0 Ir}{2\pi a^2}$, $r = \frac{2a}{3}$ par $B = \frac{\mu_0 I}{3\pi a}$. **Ans. 3**

42. $\chi \propto \frac{1}{T}$, T aadha to χ doguna $= 0.015$. **Ans. 3**

43. $F = (2e)vB = 3.2 \times 10^{-19} \times 4 \times 2 = 25.6 \times 10^{-19}$ N. **Ans. 2**

44. Wires ke beech fields add hote hain. Bahar (A aur C) par cancel ho sakte hain. **Ans. 3**

45. Radial lines ka contribution 0. Dono arcs ki direction opposite: $\frac{\mu_0 I}{4R} - \frac{\mu_0 I}{8R} = \frac{\mu_0 I}{8R}$. **Ans. 3**

CHEMISTRY

46. Halogen ns^2np^5. **Ans. 2**

47. Metallic character period mein ghatta hai, group mein badhta hai. **Ans. 1**

48. Radius $\propto \frac{1}{Z_{eff}}$. **Ans. 4**

49. IE_1 gaseous atom se, $X(g) \rightarrow X^+(g) + e^-$. **Ans. 1**

50. O ka EGE O > S anomaly ki wajah se kam hai, S sabse

negative. **Ans. 2**

51. Metalloids p-block mein. **Ans. 2**

52. $1s^2 2s^2 2p^6 3s^1$ = Na, ye metal hai, non-metal nahi. **Ans. 1**

53. BaO sabse basic. **Ans. 4**

54. 3rd period mein 8 elements. **Ans. 4**

55. $IE_1 < IE_2 < IE_3$. **Ans. 3**

56. Uut $Z = 113$, p-block. **Ans. 2**

57. N_2O<NO<N_2O_3<NO_2 hona chahiye, isliye N_2O<NO_2<N_2O_3 galat. **Ans. 2**

58. Be > B > Li > Na. **Ans. 1**

59. N ki electron affinity positive (least). **Ans. 2**

60. F, Cl, Br triad nahi hai (Cl ka atomic mass F aur Br ka mean nahi). **Ans. 3**

61. $(n-1)d^6 ns^2$ sabse chhota Z = Fe ($3d^6 4s^2$), 4th period. **Ans. 3**

62. $3d^2 4s^2$ (Ti), d-block. **Ans. 4**

63. 119 = un-un-enn = Uue. **Ans. 4**

64. M^{2+} = [Ar]$3d^1$ to M = [Ar]$3d^1 4s^2$ = Sc, period 4, group 3. **Ans. 3**

65. Cs sabse electropositive. **Ans. 2**

66. IE_2: Li > B > Be. **Ans. 2**

67. Be 899.5, Ne 2080, He 2372, N 1402.3. **Ans. 3**

68. Li, Na, K (valence mein 1 e^-). **Ans. 4**

69. N > O > F (period mein radius ghatta hai). **Ans. 4**

70. O^{2-}>O^->O sahi. Baaki galat. **Ans. 3**

71. Pd(46): gr 10, per 5. Ce(58): f-block. Ba(56): gr 2, per 6 (sahi). Mo(42): gr 6, per 5 (sahi). Sirf C, D. **Ans. 2**

72. $O^- + e^- \rightarrow O^{2-}$ endothermic (repulsion). **Ans. 2**

73. B–Si diagonal pair. **Ans. 4**

74. Li-s, Fe-d, P-p, U-f. **Ans. 1**

75. 37-Rb (s), 78-Pt (d), 52-Te (p), 65-Tb (f). **Ans. 4**

76. Cs: [Xe]$6s^1$. **Ans. 3**

77. S, p-block. **Ans. 2**

78. 4th period Kr par khatam. **Ans. 4**

79. N > C > B. **Ans. 2**

80. F electron easily nahi chhodta. **Ans. 2**

81. Ba sabse kam IE. **Ans. 3**

82. P > S sahi (half-filled), Na < Mg sahi. **Ans. 3**

83. $1s^2 2s^2 2p^6 3s^2 3p^4$ = S. **Ans. 4**

84. F ka radius sabse chhota. **Ans. 3**

85. EN: N(3.0) > C(2.5) > P(2.1) > Si(1.8). **Ans. 3**

86. $Z = 55$ Cs, s-block. **Ans. 1**

87. Cl > F > Br > I. **Ans. 4**

88. P^{3+}>P^{5+} (charge badhne par size ghatta hai). **Ans. 4**

89. Eka-Si = Ge, Eka-Al = Ga, Eka-Mn = Tc, teeno sahi. **Ans. 2**

90. IE_1 kam, IE_2 bahut zyada (noble gas core toot-ta hai). **Ans. 4**

BIOLOGY

91. Pill gonadotropins (FSH/LH) ka release rokti hai. **Ans. 3**

92. Sporogenous tissue ki har cell microspore mother cell (PMC) ban sakti hai. **Ans. 3**

93. Figure *Papaver* ka multicarpellary syncarpous pistil hai. **Ans. 2**

94. Dono sahi (orchid ke seeds tiny, date palm seed 2000 saal). **Ans. 3**

95. GIFT: A (donor ovum fallopian tube mein) aur C sahi. B IUI hai, D ZIFT hai. **Ans. 4**

96. Insect-pollinated mein pollen sticky hote hain, to non-sticky unse associated nahi. **Ans. 4**

97. Seed storage ke liye dehydration aur dormancy. **Ans. 1**

98. Pollen-pistil interaction: stigma par deposition se pollen tube ke ovule mein jaane tak. **Ans. 4**

99. *Vallisneria*, *Zostera* aur sea-grasses, sab. **Ans. 4**

100. GIFT test tube baby programme (IVF-ET) ka hissa nahi hai. **Ans. 4**

101. Rosaceae aur Solanaceae mein pollen viability mahino tak. **Ans. 4**

102. Mature embryo sac 7-celled, 8-nucleate. **Ans. 1**

103. Coconut water free-nuclear endosperm hai. **Ans. 2**

104. Cleistogamous flowers hamesha autogamy dikhate hain. **Ans. 3**

105. "Quality of life worsen karna" importance nahi dikhata. **Ans. 3**

106. A, C, D sahi. B galat (outer wall sporopollenin se bani hai,

pectin se nahi). **Ans. 1**

107. A, B sahi. C mein wall do-layered hai, D mein exine sporopollenin ki hoti hai. **Ans. 2**

108. Pollen-pistil ka chemical dialogue decide karta hai. **Ans. 3**

109. Egg apparatus micropylar end par, 3 cells (1 egg + 2 synergids). **Ans. 3**

110. Barrier ke saath chemical use karne se efficiency badhti hai. **Ans. 1**

111. Apomixis asexual reproduction hai, sexual nahi. **Ans. 3**

112. A sahi, R sahi, par R, A ki explanation nahi. **Ans. 4**

113. A sahi, B sahi, C galat (4 theca nahi, 2 theca per lobe), D sahi. **Ans. 3**

114. Dono sahi, par R, A ki explanation nahi. **Ans. 4**

115. Lagatar cross-pollination se nahi, balki lagatar self-pollination se inbreeding depression hota hai. **Ans. 4**

116. Dono sahi, par R, A ki explanation nahi. **Ans. 4**

117. A-IV (antipodals), B-III (egg apparatus), C-II (filiform apparatus), D-I (central cell). **Ans. 2**

118. Dono sahi, R sahi explanation. **Ans. 3**

119. Proembryo → Globular → Heart-shaped → Mature. **Ans. 2**

120. Syngamy se diploid embryo, triple fusion se triploid endosperm. **Ans. 1**

121. Dono sahi, R sahi explanation. **Ans. 3**

122. Rice aur wheat ke pollen release ke 30 min ke andar viability kho dete hain. **Ans. 3**

123. Self-incompatibility genetically controlled mechanism hai. **Ans. 1**

124. A = thalamus, B = achene. **Ans. 3**

125. *Zostera* ke female flowers pollination ke dauran submerged rehte hain. **Ans. 3**

126. Perisperm chukandar aur kali mirch mein. **Ans. 3**

127. *Zostera* ke pollen ribbon-like. **Ans. 2**

128. Food storage endosperm mein hota hai, figure mein C. **Ans. 3**

129. 60% se zyada angiosperms mein pollen 2-celled stage par jhadte hain, 3-celled par nahi. **Ans. 2**

130. Emasculation mein anthers hataye jaate hain, ovary nahi. **Ans. 2**

131. Cotyledon ke upar ka part epicotyl hai, hypocotyl nahi. **Ans. 3**

132. Dono sahi. **Ans. 3**

133. Dioecious condition autogamy aur geitonogamy dono rokti hai. **Ans. 2**

134. Embryo micropylar end par banta hai. **Ans. 1**

135. Epidermis → Endothecium → Middle layers → Tapetum. **Ans. 3**

136. India ne sabse pehle national level par RCH programme shuru kiya. **Ans. 2**

137. OCP STD se nahi bachati. **Ans. 3**

138. Oral pills kam side effect wali aur well accepted. **Ans. 3**

139. Cu ions sperm motility aur fertilising capacity ko suppress karte hain (B, D). **Ans. 4**

140. Rubber sheath = condom. **Ans. 2**

141. Haploid: egg, synergids, antipodals. **Ans. 4**

142. Khujli STD ka early symptom. **Ans. 4**

143. Emergency contraception unprotected coitus ke 72 ghante ke andar. **Ans. 3**

144. LNG-20 hormone releasing IUD hai, non-medicated nahi. **Ans. 3**

145. Wind pollination mein ribbon-like pollen nahi hote (ye *Zostera* mein hota hai). **Ans. 2**

146. Teeno natural methods hain. **Ans. 4**

147. Figure Copper-T ki hai. **Ans. 3**

148. "Hum Do Hamare Do" = chhota parivar, do bachche. **Ans. 2**

149. Saheli non-steroidal hai aur saptahik (A, B). **Ans. 4**

150. Kela parthenocarpic hai. **Ans. 4**

151. Statement I sahi. II galat (MTP voluntary hota hai, involuntary nahi). **Ans. 2**

152. I sahi. II galat (pills ovulation rokti hain). **Ans. 2**

153. Surgical methods gamete transport rokti hain, gamete formation nahi. **Ans. 2**

154. Diaphragm cervix se sperm entry rokta hai. **Ans. 1**

155. *Orobanche* aur *Striga* parasitic. **Ans. 3**

156. MTP intentional hota hai aur induced abortion bhi kehlata hai. **Ans. 4**

157. Diaphragm, cap aur vault cervix cover karte hain aur reusable hain. **Ans. 4**

158. Sirf III aur IV sahi, count = 2. **Ans. 2**

159. Ye ZIFT ki characteristics hain. **Ans. 2**

160. MMC: 1 meiosis + 3 mitosis. **Ans. 2**

161. Periodic abstinence cycle ke 10 se 17 din. **Ans. 3**

162. RCH-IV, CDRI-II, STD-I, WHO-III. **Ans. 4**

163. Lactational amenorrhea parturition ke baad max 6 mahine. **Ans. 3**

164. Charon factors sahi. **Ans. 4**

165. IUT-IV, IUI-III, ZIFT-I, ICSI-II. **Ans. 4**

166. Gonorrhoea, Syphilis, Chlamydiasis, Trichomoniasis curable hain (4). Hepatitis-B, herpes, HIV, warts viral hain. **Ans. 2**

167. OCP myometrium ki thickness nahi badhati. **Ans. 2**

168. Cu-T doctor ya nurse yoni se lagate hain. **Ans. 2**

169. Rhythm method sabse zyada fail hota hai. **Ans. 4**

170. Vasectomy mein vas deferens kata jata hai, testis ka hissa nahi. **Ans. 3**

171. VD, RTI, STI, teeno terms. **Ans. 4**

172. Lactation mein gonadotropins suppress hote hain. **Ans. 1**

173. MTP 12 weeks tak safe. **Ans. 1**

174. IUDs delay aur spacing ke liye ideal. **Ans. 3**

175. Coitus interruptus. **Ans. 2**

176. A, B, C sahi. D galat (females ke liye 18 aur males ke liye 21 saal hai). **Ans. 1**

177. A aur R dono sahi, R sahi explanation. **Ans. 3**

178. Dono statement sahi. **Ans. 1**

179. ICSI mein sperm seedha ovum mein inject hota hai. **Ans. 2**

180. Cervical cap female tract mein lagti hai (I galat), II sahi. **Ans. 3**